Constructive KAM Methods for Quasiperiodic Lagrangian Tori: Algorithms and Applications

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Hamiltonian Systems with Quasiperiodic Time Dependence

We consider non-autonomous Hamiltonian systems

$$h(q, p, \alpha t), \qquad \alpha = (\alpha_1, \ldots, \alpha_\ell) \in \mathbb{R}^\ell.$$

Quasiperiodic time dependence

Introduce external phases

$$\varphi = \alpha t + \varphi_0, \qquad \varphi \in \mathbb{T}^\ell,$$

satisfying

$$\dot{\varphi} = \alpha.$$

The dynamics can then be written as the autonomous vector field

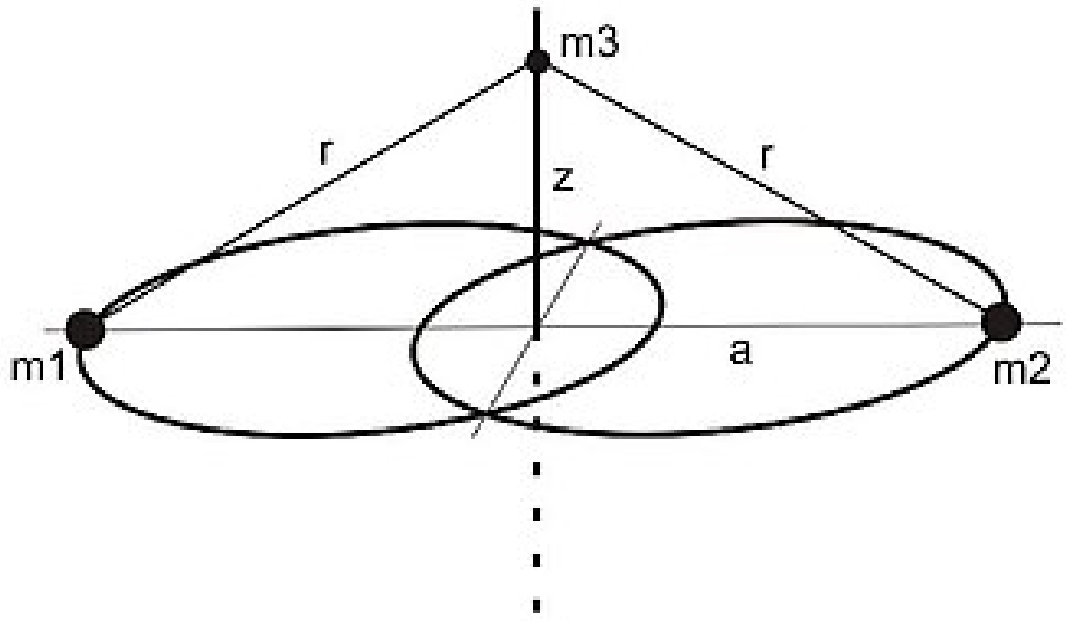
$$\widetilde{Z}(z, \varphi) = \begin{pmatrix} Z_h(z, \varphi) \\ \alpha \end{pmatrix},$$

where $z = (q, p)$

Examples from Physics

These systems appear naturally in various physical models:

- **Celestial Mechanics**

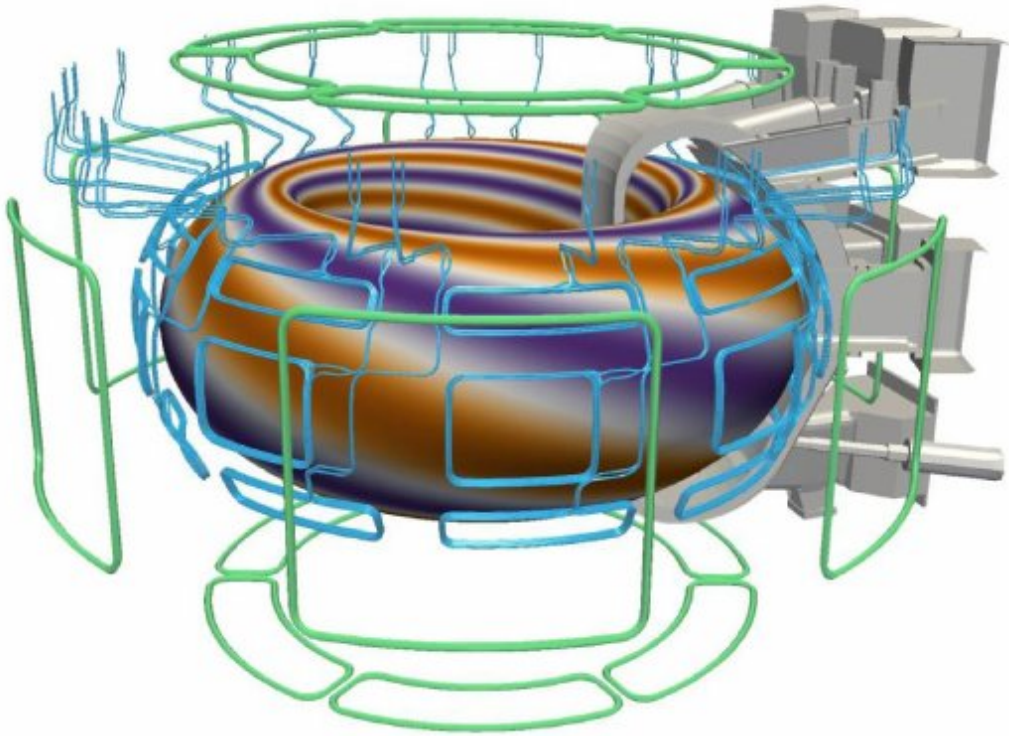


Configuration of the Sitnikov problem

Examples from Physics

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- **Celestial Mechanics**
- **Plasma Physics**

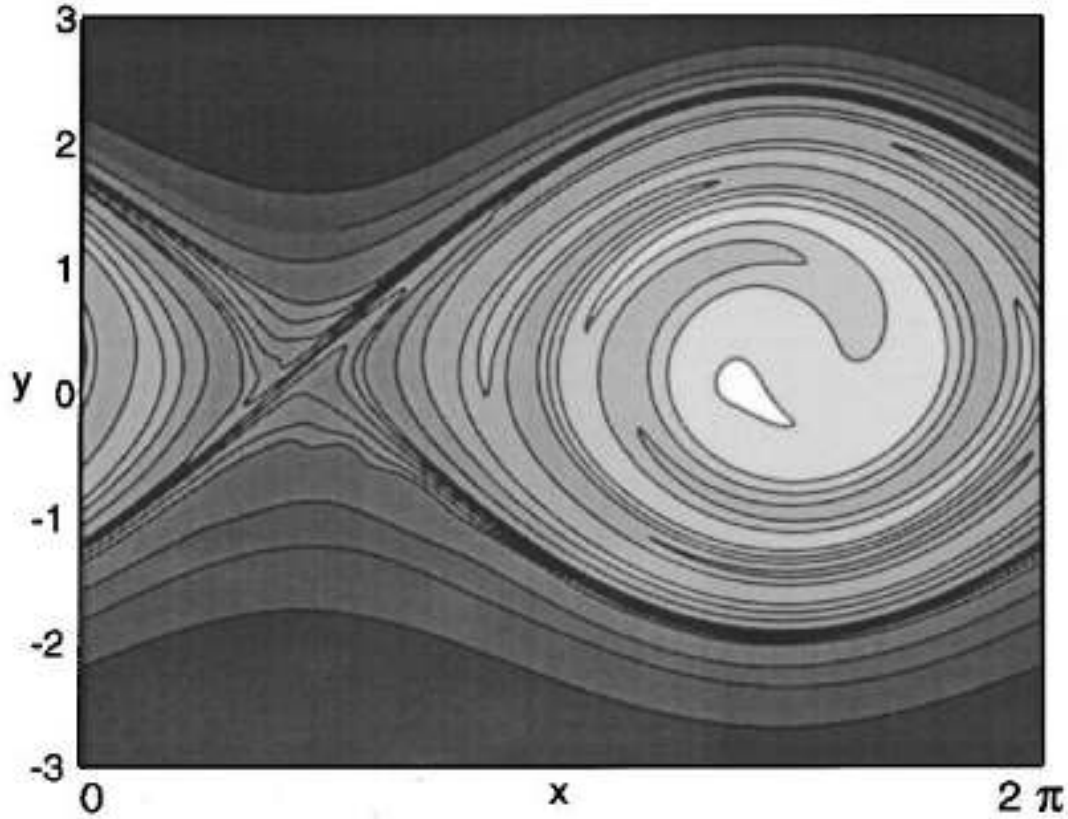


ITER Tokamak for plasma confinement.
Source: `https://www.iter.org/`

Examples from Physics

These systems appear naturally in various physical models:

- **Celestial Mechanics**
- **Plasma Physics**
- **Fluid Mechanics**



Flow instability and vortex formation
Source: Diego del-Castillo-Negrete

The Setting

We consider an exact symplectic structure $\boldsymbol{\omega}$, a Riemannian metric $\boldsymbol{g}$ on the open set $\mathcal{M} \subset \mathbb{R}^{2n}$, and a linear isomorphism $\boldsymbol{J}$, between these objects, $\omega_z(J_z u, v) = g_z(u, v)$, $\forall u, v \in T_z\mathcal{M}$. Their matrix representations are

$$\Omega^\top = -\Omega\,, \qquad G^\top = G\,, \qquad J^\top \Omega = G\,.$$

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The quasi-periodic Hamiltonian vector field is

$$\tilde{Z}(z,\varphi) = \begin{cases} Z_h(z,\varphi) := \Omega(z)^{-1} \mathrm{D}_z h(z,\varphi)^\top\,, \\ \alpha\,, \end{cases}$$

$\alpha \in \mathbb{R}^\ell$ is **external frequency vector**, and $\varphi \in \mathbb{T}^\ell$ (if $\ell = 1$, φ plays a role similar to time).

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$\alpha \in \mathbb{R}^\ell$ is **external frequency vector**, and $\varphi \in \mathbb{T}^\ell$ (if $\ell = 1$, φ plays a role similar to time).
The flow on $\mathcal{M} \times \mathbb{T}^\ell$ takes the form

$$\tilde{\phi}_t(z,\varphi) = \begin{pmatrix} \phi_t(z,\varphi) \\ \varphi + \alpha t \end{pmatrix}, \text{with}$$

where

$$\frac{\partial \phi_t}{\partial t}(z,\varphi) = Z_h(\phi_t(z,\varphi), \varphi + \alpha t), \text{and } \phi_0(z,\varphi) = z.$$

Invariant tori

We look for a parametrization $\tilde{K} : \mathbb{T}^n \times \mathbb{T}^\ell \to \mathbb{R}^n \times \mathbb{R}^\ell$, of the form

$$\tilde{K}(\theta, \varphi) = \begin{pmatrix} K(\theta, \varphi) \\ \varphi \end{pmatrix},$$

where $K : \mathbb{T}^n \times \mathbb{T}^\ell \to \mathbb{R}^n$ satisfies the **invariance condition**

$$\phi_t\big(K(\theta, \varphi), \varphi\big) = K(\theta + t\omega, \varphi + t\alpha).$$

Invariant tori

Differentiating with respect to t at $t = 0$ yields the **infinitesimal invariance equation**:

Infinitesimal invariance

$$Z_h\big(K(\theta,\varphi),\varphi\big) \;=\; -\mathfrak{L}_{\omega,\alpha}K(\theta,\varphi),$$

where the linear operator $\mathfrak{L}_{\omega,\alpha}$ is defined as

$$\mathfrak{L}_{\omega,\alpha}(\cdot) \;:=\; D_\theta(\cdot)\,\omega \;+\; D_\varphi(\cdot)\,\alpha.$$

The frequency vector $(\omega,\alpha) \in \mathcal{D}_{\gamma,\tau}$ is **Diophantine**.

$$\mathcal{D}_{\gamma,\tau} = \left\{(\omega,\alpha)\in\mathbb{R}^d\times\mathbb{R}^\ell \;:\; \big|k_1\cdot\omega + k_2\cdot\alpha\big| \;\geq\; \frac{\gamma}{\|(k_1,k_2)\|_1^\tau},\; \forall (k_1,k_2)\in\mathbb{Z}^d\times\mathbb{Z}^\ell\right.$$

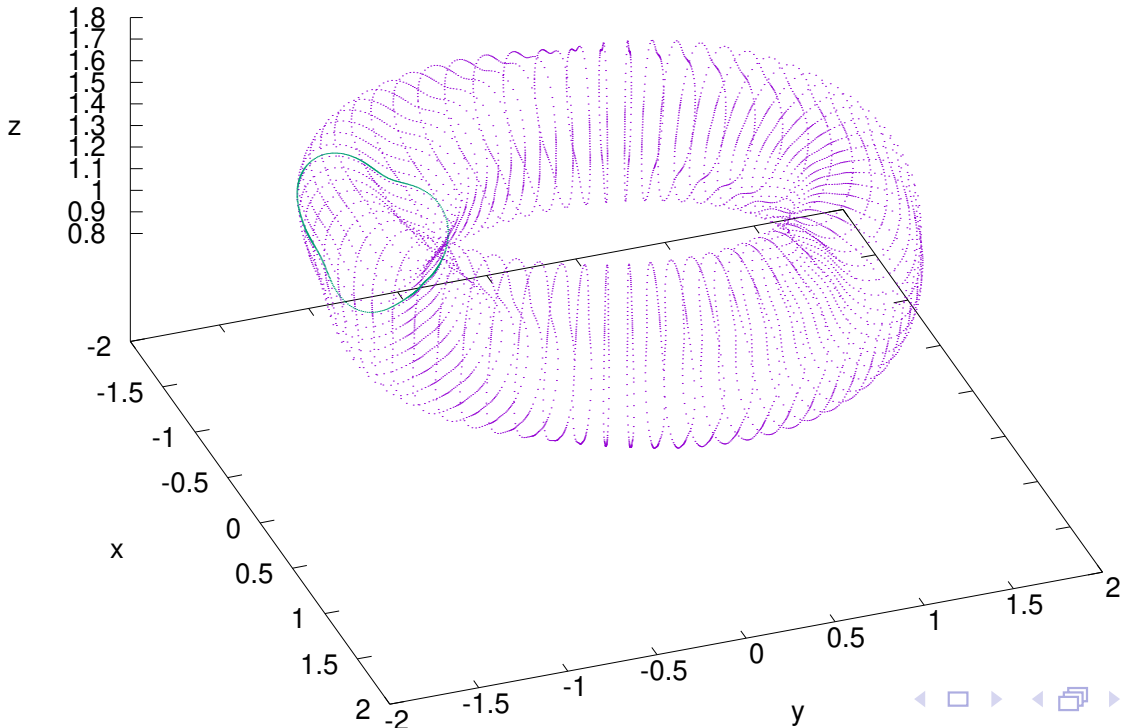
Fiberwise structure

Geometrically, the parametrization

$$\tilde{K}(\theta, \varphi) = \begin{pmatrix} K(\theta, \varphi) \\ \varphi \end{pmatrix}$$

defines a **fibration** of the phase space:

- For each fixed $\varphi \in \mathbb{T}^\ell$, $K(\theta, \varphi)$ parametrizes an invariant torus.
- As φ varies, we obtain a family of tori, one for each angle φ.



Symplectic frame (I)

Recall that we already know:

- The parametrization $K(\theta, \varphi)$ of the invariant torus,
- The symplectic matrix Ω, the metric G, and the almost complex structure J,

We now construct tangent and normal frames associated with the torus.

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Tangent and normal frames:

$$L(\theta, \varphi) := D_\theta K(\theta, \varphi)\,, \tag{1}$$

$$N(\theta, \varphi) := L(\theta, \varphi) A(\theta, \varphi) + N^0(\theta, \varphi) B(\theta, \varphi)\,, \tag{2}$$

where the auxiliary quantities are:

$$N^0(\theta, \varphi) := J(K(\theta, \varphi))\, L(\theta, \varphi)\,,$$

$$B(\theta, \varphi) := \left(L(\theta, \varphi)^\top G(K(\theta, \varphi)) L(\theta, \varphi)\right)^{-1},$$

$$A(\theta, \varphi) := -\frac{1}{2}\, B(\theta, \varphi)^\top\, N^0(\theta, \varphi)^\top \Omega(K(\theta, \varphi))\, N^0(\theta, \varphi)\, B(\theta, \varphi)\,.$$

Symplectic frame (II)

Symplectic property

The frame

$$P(\theta,\varphi) = \begin{pmatrix} L(\theta,\varphi) & N(\theta,\varphi) \end{pmatrix}$$

is **symplectic** for the invariant torus:

$$P(\theta,\varphi)^{\top}\,\Omega(K(\theta,\varphi))\,P(\theta,\varphi) = \Omega_0,$$

where Ω_0 is the standard symplectic matrix.

The quasi-Newton method (I/III): Setup

Following the a-posteriori approach to KAM theory (see Calleja, Haro & Porras, 2025, for details), we start from an **approximate solution** $K(\theta, \varphi)$.

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Invariance error

Define the error functional

$$E(\theta, \varphi) \;:=\; Z_h\big(K(\theta, \varphi), \varphi\big) \;+\; \mathfrak{L}_{\omega,\alpha} K(\theta, \varphi),$$

where (ω, α) is a **Diophantine frequency vector**.

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Goal

Find a correction $\Delta K(\theta, \varphi)$ such that

$$\bar{K}(\theta, \varphi) \;:=\; K(\theta, \varphi) + \Delta K(\theta, \varphi)$$

yields a smaller invariance error.

The quasi-Newton method (II/III): Linearization

Correction ansatz:

$$\Delta K(\theta, \varphi) \;=\; P(\theta, \varphi)\, \xi(\theta, \varphi),$$

where $P(\theta, \varphi) = \begin{pmatrix} L(\theta, \varphi) & N(\theta, \varphi) \end{pmatrix}$ is the symplectic frame, and $\xi : \mathbb{T}^n \times \mathbb{T}^\ell \to \mathbb{R}^{2n}$ is unknown.

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Linearized equation

Expanding around the approximate torus and keeping only first-order terms gives

$$D_z Z_h(K(\theta,\varphi),\varphi)\,\Delta K(\theta,\varphi) \;+\; \mathfrak{L}_{\omega,\alpha}\Delta K(\theta,\varphi) \;=\; -E(\theta,\varphi).$$

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Cohomological equation in Fourier space

Substituting $\Delta K = P\xi$ and projecting with $P^\top\Omega$ yields

$$\Lambda(\theta,\varphi)\,\xi(\theta,\varphi) \;+\; \mathfrak{L}_{\omega,\alpha}\xi(\theta,\varphi) \;=\; -P(\theta,\varphi)^\top\Omega(K(\theta,\varphi))\,E(\theta,\varphi) \;=:\; \eta(\theta,\varphi),$$

where

$$\Lambda(\theta,\varphi) = \begin{pmatrix} 0_n & T(\theta,\varphi) \\ 0_n & 0_n \end{pmatrix}, \qquad T = N^\top\Omega(K)\Big(D_z Z_h(K,\varphi)\,N \;+\; \mathfrak{L}_{\omega,\alpha}N\Big).$$

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$$\mathfrak{L}_{\omega,\alpha}\xi^N(\theta,\varphi) = \eta^N(\theta,\varphi),$$

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Formal Fourier solution

For $k, j \neq 0$, the Fourier coefficients satisfy

$$\hat{\xi}_{kj} = \frac{\hat{\eta}_{kj}}{\mathbf{i}\,2\pi\,(k\omega + j\alpha)},$$

while the zero mode $\hat{\xi}_{00} \in \mathbb{R}$ is **free**.

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Convergence

By applying this quasi-Newton method **iteratively**, we obtain a sequence of approximations converging to a truly invariant torus. The convergence is proved in a suitable Banach space of analytic functions.

KAM algorithm

Input: Given K, we compute.

1. $L(\theta,\varphi) = D_\theta K(\theta,\varphi)$
2. $B(\theta,\varphi) = (L(\theta,\varphi)^\top G(K(\theta,\varphi))L(\theta,\varphi))^{-1}$
3. $N^0(\theta,\varphi) = J(K(\theta,\varphi))L(\theta,\varphi)$
4. $A(\theta,\varphi) = -\frac{1}{2}B(\theta,\varphi)^\top N(\theta,\varphi)^{0,\top}\Omega(K(\theta,\varphi))N(\theta,\varphi)^0 B(\theta,\varphi)$
5. $N(\theta,\varphi) = L(\theta,\varphi)A(\theta,\varphi) + N^0(\theta,\varphi)B(\theta,\varphi)$
6. $T(\theta,\varphi) = N(\theta,\varphi)^\top\Omega(K(\theta,\varphi))\left(\mathrm{D}_z Z_h(K(\theta,\varphi),\varphi)N(\theta,\varphi) + \mathfrak{L}_{\omega,\alpha}N(\theta,\varphi)\right)$
7. $P(\theta,\varphi) = \big(L(\theta,\varphi) \mid N(\theta,\varphi)\big)$
8. $E(\theta,\varphi) = Z_h(K(\theta,\varphi),\varphi) + \mathfrak{L}_{\omega,\alpha}K(\theta,\varphi)$
9. $\eta(\theta,\varphi) = -P(\theta,\varphi)^\top\Omega(K(\theta,\varphi))E(\theta,\varphi)$
10. Solve the cohomological equations for ξ^N and ξ^L.
11. Update the torus: $K(\theta,\varphi) \leftarrow K(\theta,\varphi) + P(\theta,\varphi)\xi(\theta,\varphi)$.

KAM Theorem for Quasi-Periodic Hamiltonian Systems

We consider an exact symplectic structure $\boldsymbol{\omega} = \mathrm{d}\boldsymbol{\alpha}$ and a Riemannian metric $\boldsymbol{g}$ on the open set $\mathcal{M} \subset \mathbb{R}^{2n}$. Let $h : \mathcal{M} \times \mathbb{T}^\ell \to \mathbb{R}$ be an analytic Hamiltonian function that depends quasi-periodically on time, with $1 \leq \ell \leq n$. Moreover, let $\mathcal{B} \subset \mathbb{C}^{2n}$, $\mathbb{T}^\ell_\varrho$ be a complex strip of width $\varrho > 0$, and we assume that h extends analytically to $\mathcal{B} \times \mathbb{T}^\ell_\varrho$.

For $\varrho > \rho > 0$, let $K : \mathbb{T}^n_\rho \times \mathbb{T}^\ell_\rho \to \mathcal{M}$ be an analytic function in the interior and continuous on the boundary, which parameterizes an approximately invariant Lagrangian torus with frequency vector $(\omega, \alpha) \in \mathbb{R}^n \times \mathbb{R}^\ell$.

The tangent frame $L : \mathbb{T}^n \times \mathbb{T}^\ell \to \mathbb{R}^{2n \times n}$ is given by:

$$L(\theta, \varphi) = \mathrm{D}_\theta K(\theta, \varphi).$$

We assume that the following hypotheses hold:

Hypotheses

The global objects extend analytically to $\mathcal{B} \subset \mathbb{C}^{2n}$. For the structures $\boldsymbol{\omega}, \boldsymbol{g}, \boldsymbol{J}$ on $\mathcal{B}$, the matrix functions $\Omega, G, J : \mathcal{B} \to \mathbb{C}^{2n\times 2n}$ (with $J(z) = -\Omega(z)^{-1}G(z)$) satisfy:

$$\|\Omega\|_{\mathcal{B}} \leq c_{\Omega,0}\,, \qquad \|\mathrm{D}_z\Omega\|_{\mathcal{B}} \leq c_{\Omega,1}\,,$$

$$\|G\|_{\mathcal{B}} \leq c_{G,0}\,, \qquad \|\mathrm{D}_z G\|_{\mathcal{B}} \leq c_{G,1}\,, \qquad \|\mathrm{D}_z^2 G\|_{\mathcal{B}} \leq c_{G,2}\,,$$

$$\|J\|_{\mathcal{B}} \leq c_{J,0}\,, \qquad \|\mathrm{D}_z J\|_{\mathcal{B}} \leq c_{J,1}\,, \qquad \|\mathrm{D}_z^2 J\|_{\mathcal{B}} \leq c_{J,2}\,,$$

$$\|J^\top\|_{\mathcal{B}} \leq c_{J^\top,0}\,, \qquad \|\mathrm{D}_z J^\top\|_{\mathcal{B}} \leq c_{J^\top,1}\,,$$

$$\|J^{-1}\|_{\mathcal{B}} \leq c_{J^{-1},0}\,,$$

$$\|(J^{-1})^\top\|_{\mathcal{B}} \leq c_{J^{-\top},0}\,.$$

$\|\cdot\|_{\mathcal{B}}$: supremum norm over $\mathcal{B}$.

Hypotheses

For the Hamiltonian $h : \mathcal{B} \times \mathbb{T}^\ell_\varrho \to \mathbb{C}$ and its vector field $Z_h : \mathcal{B} \times \mathbb{T}^\ell_\varrho \to \mathbb{C}^{2n}$, defined by $Z_h(z,\varphi) := \Omega(z)^{-1} \left(\mathrm{D}_z h(z,\varphi)\right)^\top$, and the operator $T_h : \mathcal{B} \times \mathbb{T}^\ell_\varrho \to \mathbb{C}^{2n}$ the following bounds hold:

$$\|\mathrm{D}_z h\|_{\mathcal{B}\times\mathbb{T}^\ell_\varrho} \leq c_{h,1}\,,$$

$$\|Z_h\|_{\mathcal{B}\times\mathbb{T}^\ell_\varrho} \leq c_{Z_h,0}\,, \qquad \|\mathrm{D}_z Z_h\|_{\mathcal{B}\times\mathbb{T}^\ell_\varrho} \leq c_{Z_h,1}\,,$$

$$\|T_h\|_{\mathcal{B}\times\mathbb{T}^\ell_\varrho} \leq c_{T_h,0}\,, \qquad \|\mathrm{D}_z T_h\|_{\mathcal{B}\times\mathbb{T}^\ell_\varrho} \leq c_{T_h,1}\,,$$

$$\|T_h^\top\|_{\mathcal{B}\times\mathbb{T}^\ell_\varrho} \leq c_{T_h^\top,0}\,, \qquad \|\mathrm{D}_z T_h^\top\|_{\mathcal{B}\times\mathbb{T}^\ell_\varrho} \leq c_{T_h^\top,1}\,,$$

$$\|\mathrm{D}_z^2 Z_h\|_{\mathcal{B}\times\mathbb{T}^\ell_\varrho} \leq c_{Z_h,2}\,, \qquad \|\mathrm{D}_z Z_h^\top\|_{\mathcal{B}\times\mathbb{T}^\ell_\varrho} \leq c_{Z_h^\top,1}\,.$$

Hypotheses

The parametrization K is analytic on $\mathbb{T}^n_\rho \times \mathbb{T}^\ell_\rho$ and

$$\operatorname{dist}(K(\mathbb{T}^n_\rho \times \mathbb{T}^\ell_\rho), \partial(\mathcal{B} \times \mathbb{T}^\ell_\varrho)) > 0.$$

Moreover,

$$\|L\|_\rho < \sigma_L, \qquad \|L^\top\|_\rho < \sigma_{L^\top}.$$

$L(\theta, \varphi)$ has maximal rank for all (θ, φ). There exists σ_B such that

$$\|B\|_\rho < \sigma_B,$$

where $B(\theta, \varphi)$ is the transversality condition.
There exists σ_T such that

$$|\langle T \rangle^{-1}| < \sigma_T,$$

where $T(\theta, \varphi)$ is the torsion.
The frequency vector $(\omega, \alpha) \in \mathcal{D}_{\gamma,\tau}$ is Diophantine for certain $\gamma > 0$ and $\tau \geq n + \ell - 1$.

Conclusion of the Theorem

Under the above hypotheses, for each $0 < \rho_\infty < \rho$ there exists a constant $\mathfrak{C}_1$ such that, if the invariance error

$$E(\theta,\varphi) = Z_h(K(\theta,\varphi),\varphi) - \mathrm{D}_\theta K(\theta,\varphi)\omega - \mathrm{D}_\varphi K(\theta,\varphi)\alpha,$$

satisfies

$$\frac{\mathfrak{C}_1 \|E\|_\rho}{\gamma^4 \rho^{4\tau}} < 1,$$

then there exists an invariant torus $\mathcal{K}_\infty = K_\infty(\mathbb{T}^n \times \mathbb{T}^\ell)$ with frequencies (ω,α). K_∞ is analytic on $\mathbb{T}^n_{\rho_\infty} \times \mathbb{T}^\ell_{\rho_\infty}$ and continuous on the boundary, with

$$\|\mathrm{D}_\theta K_\infty\|_{\rho_\infty} < \sigma_L, \quad \|(\mathrm{D}_\theta K_\infty)^\top\|_{\rho_\infty} < \sigma_{L^\top},$$

$$\mathrm{dist}(K_\infty(\mathbb{T}^n_{\rho_\infty} \times \mathbb{T}^\ell_{\rho_\infty}), \partial(\mathcal{B} \times \mathbb{T}^\ell_\varrho)) > 0.$$

Moreover, there exists $\mathfrak{C}_2$ such that

$$\|K_\infty - K\|_{\rho_\infty} < \frac{\mathfrak{C}_2 \|E\|_\rho}{\gamma^2 \rho^{2\tau}}.$$

On the KAM constants

The constants $\mathfrak{C}_1$ and $\mathfrak{C}_2$ are given explicitly in the Appendix.

Table 1: Constants introduced in Section 4. Constants with * in the label correspond to **Case III**.

Object	Constant	Label	Result
$\mathfrak{L}_\omega \Omega_L$	$C_{\mathfrak{L}\Omega_L} = 2nc_{\Omega,0}\sigma_L + \sigma_{L^\top} c_{\Omega,1}\sigma_L \delta + n\sigma_{L^\top} c_{\Omega,0}$	(4.4)	Lemma 4.2
Ω_L	$C_{\Omega_L} = c_R C_{\mathfrak{L}\Omega_L}$	(4.5)	Lemma 4.2
L	$C_L = \sigma_L$	(4.9)	Lemma 4.4
$L^\top$	$C_{L^\top} = \sigma_{L^\top}$	(4.10)	Lemma 4.4
G_L	$C_{G_L} = C_{L^\top} c_{G,0} C_L$	(4.11)	Lemma 4.4
N^0	$C_{N^0} = c_{J,0} C_L$	(4.15)	Lemma 4.5
$(N^0)^\top$	$C_{N^{0,\top}} = C_{L^\top} c_{J^\top,0}$	(4.16)	Lemma 4.5
$\hat{N}$	$C_{\hat{N}} = C_{N^0} \sigma_B$	(4.17)	Lemma 4.5
$\hat{N}^\top$	$C_{\hat{N}^\top} = \sigma_B C_{N^{0,\top}}$	(4.18)	Lemma 4.5
A	$C_A = \frac{1}{2} C_{\hat{N}^\top} c_{\Omega,0} C_{\hat{N}}$, $^*C_A = 0$	(4.19)*	Lemma 4.5
N	$C_N = C_L C_A + C_{\hat{N}}$, $^*C_N = C_{\hat{N}}$	(4.20)	Lemma 4.5
$N^\top$	$C_{N^\top} = C_A C_{L^\top} + C_{\hat{N}^\top}$, $^*C_{N^\top} = C_{\hat{N}^\top}$	(4.21)	Lemma 4.5
E_{sym}	$C_{\mathrm{sym}} = (1 + C_A)\max\{1, C_A\} C_{\Omega_L}$, $^*C_{\mathrm{sym}} = C_{\Omega_L} \max\{1, (\sigma_B)^2\}$	(4.25)	Lemma 4.5
T	$C_T = \frac{1}{2}(C_{\hat{N}^\top} c_{T_h,0} C_{\hat{N}} + C_{\hat{N}^\top} c_{T_h^\top,0} C_{\hat{N}}) + C_{\hat{N}^\top} c_{T_h^\top,0} C_N + C_{N^\top} c_{T_h,0} C_{\hat{N}}$ $^*C_T = C_{N^\top} c_{T_h,0} C_N$	(4.26)	Lemma 4.7
T_E	$C_{T_E} = c_{\Omega,1}\delta + c_{\Omega,0} c_{J,1} c_{J^{-1},0}\delta + 2nc_{J^\top,0} C_L \sigma_B C_{\Omega,0}^2 + nc_{J^\top,0}\sigma_B C_{L^\top} c_{\Omega,0}^2$	(4.44)	Lemma 4.10
$T_E^\top$	$C_{T_E^\top} = c_{\Omega,1}\delta + c_{J^{-\top},0} c_{J^\top,1} c_{\Omega,0}\delta + n\sigma_B C_{L^\top} c_{J,0} c_{\Omega,0}^2 + 2nC_L \sigma_B c_{J,0} c_{\Omega,0}^2$	(4.45)	Lemma 4.10
$E_{\mathfrak{L}B}$	$C_{E_{\mathfrak{L}B}} = C_{\hat{N}^\top} c_{J^{-\top},0} C_{T_E} C_{\hat{N}}$	(4.46)	Lemma 4.10
$E_{\mathfrak{L}B}^\top$	$C_{E_{\mathfrak{L}B}^\top} = C_{\hat{N}^\top} C_{T_E^\top} c_{J^{-1},0} C_{\hat{N}}$	(4.47)	Lemma 4.10
$E_{\mathfrak{L}\hat{N}}$	$C_{E_{\mathfrak{L}\hat{N}}} = \delta c_{J,1} C_L \sigma_B + nc_{J,0}\sigma_B + c_{J,0} C_L C_{E_{\mathfrak{L}B}}$	(4.48)	Lemma 4.10
$E_{\mathfrak{L}\hat{N}}^\top$	$C_{E_{\mathfrak{L}\hat{N}}^\top} = \delta\sigma_B C_{L^\top} c_{J^\top,1} + 2n\sigma_B c_{J^\top,0} + C_{E_{\mathfrak{L}B}^\top} C_{L^\top} c_{J^\top,0}$	(4.49)	Lemma 4.10
$E_{\mathfrak{L}A}$	$C_{E_{\mathfrak{L}A}} = \frac{1}{2}(C_{E_{\mathfrak{L}\hat{N}}^\top} c_{\Omega,0} C_{\hat{N}} + \delta C_{N^\top} c_{\Omega,1} C_{\hat{N}} + C_{\hat{N}^\top} c_{\Omega,0} C_{E_{\mathfrak{L}\hat{N}}})$	(4.50)	Lemma 4.10
$E_{\mathfrak{L}N}$	$C_{E_{\mathfrak{L}N}} = C_{E_{\mathfrak{L}\hat{N}}} + C_L C_{E_{\mathfrak{L}A}} + nC_A$	(4.51)	Lemma 4.10
E_T	$C_{E_T} = \frac{1}{2}(C_A C_{\Omega_L} C_{\hat{N}^\top} c_{T_h^\top,0} C_{\hat{N}} + C_A C_{\Omega_L} C_{\hat{N}^\top} c_{T_h,0} C_{\hat{N}}) + C_A C_{\Omega_L} C_{A^\top} c_{T_h,0} C_{\hat{N}} + C_A C_{\Omega_L} C_{\hat{N}^\top} c_{T_h^\top,0} C_{\hat{N}} C_L C_A + \gamma\delta^\tau C_A C_{L^\top} c_{\Omega,0} C_{E_{\mathfrak{L}N}} + \gamma\delta^\tau C_{\hat{N}^\top} c_{\Omega,0} C_{E_{\mathfrak{L}N}}$	(4.52)	Lemma 4.10
$\mathcal{X}_L$	$C_{\mathcal{X}_L} = n$	(4.62)	Lemma 4.11
$\mathcal{X}_L^\top$	$C_{\mathcal{X}_L^\top} = 2n$	(4.63)	Lemma 4.11
$\mathfrak{L}\bar{A}$	$C_{\mathfrak{L}A} = \frac{1}{2}\big(C_{\hat{N}^\top} c_{T_h^\top,0} C_{\hat{N}} + C_{\hat{N}^\top} c_{T_h^\top,0} c_{J^{-1},0} C_{\hat{N}} C_{L^\top} c_{J^\top,0} c_{\Omega,0} C_{\hat{N}} + C_{\hat{N}^\top} c_{T_h,0} C_{\hat{N}} + C_{\hat{N}^\top} c_{\Omega,0} c_{J,0} C_L C_{\hat{N}^\top} c_{J^{-\top},0} c_{T_h,0} C_{\hat{N}}\big)$ $^*C_{\mathfrak{L}A} = 0$	(4.66)	Lemma 4.11
$E_{\mathrm{red}}^{1,1}$	$C_{\mathrm{red}}^{1,1} = C_{N^\top} c_{\Omega,0} C_{\mathcal{X}_L}$	(4.68)	Lemma 4.11
$E_{\mathrm{red}}^{1,2}$	$C_{\mathrm{red}}^{1,2} = C_{E_T}$	(4.69)	Lemma 4.11
$E_{\mathrm{red}}^{2,1}$	$C_{\mathrm{red}}^{2,1} = C_{L^\top} c_{\Omega,0} C_{\mathcal{X}_L}$	(4.70)	Lemma 4.11
$E_{\mathrm{red}}^{2,2}$	$C_{\mathrm{red}}^{2,2} = (C_{L^\top} c_{\Omega,1} C_N \delta + C_{\mathcal{X}_L^\top} c_{\Omega,0} C_N + C_{\mathfrak{L}\Omega_L} C_A + \sigma_{L^\top} c_{\Omega,0}\sigma_L C_{E_{\mathfrak{L}A}})\gamma\delta^\tau + C_{\Omega_L} C_{\mathfrak{L}A}$	(4.71)	Lemma 4.11
E_{red}	$C_{\mathrm{red}} = \max\{C_{\mathrm{red}}^{1,1}\gamma\delta^\tau + C_{\mathrm{red}}^{1,2}, C_{\mathrm{red}}^{2,1}\gamma\delta^\tau + C_{\mathrm{red}}^{2,2}\}$	(4.72)	Lemma 4.11

Table 2: Constants introduced in Lemma 5.2. Given an object Z_h, we use the notation $\Delta\mathfrak{L}Z_h = \mathfrak{L}_{\omega,\alpha}\bar{Z}_h - \mathfrak{L}_{\omega,\alpha}Z_h$. Constants with * in the label are 0 in **Case III**.

Object	Constant	Label
ξ_0^N	$C_{\xi_0^N} = \sigma_T(C_{N^\top}c_{\Omega,0}\gamma\delta^\tau + c_R C_T C_{L^\top}c_{\Omega,0})$	(5.22)
ξ^N	$C_{\xi^N} = C_{\xi_0^N} + c_R C_{L^\top}c_{\Omega,0}$	(5.23)
ξ^L	$C_{\xi^L} = c_R(C_{N^\top}c_{\Omega,0}\gamma\delta^\tau + C_T C_{\xi^N})$	(5.24)
ξ	$C_\xi = \max\{C_{\xi^L}, C_{\xi^N}\gamma\delta^\tau\}$	(5.25)
ΔK	$C_{\Delta K} = C_L C_{\xi^L} + C_N C_{\xi^N}\gamma\delta^\tau$	(5.26)
$\mathfrak{L}_\omega\xi^N$	$C_{\mathfrak{L}_\xi N} = C_{L^\top}c_{\Omega,0}$	(5.29)
$\mathfrak{L}_\omega\xi^L$	$C_{\mathfrak{L}_\xi L} = C_{N^\top}c_{\Omega,0}\gamma\delta^\tau + C_T C_{\xi^N}$	(5.30)
$\mathfrak{L}_\omega\xi$	$C_{\mathfrak{L}_\xi\xi} = \max\{C_{\mathfrak{L}_\xi L}, C_{\mathfrak{L}_\xi N}\gamma\delta^\tau\}$	(5.31)
E_{lin}	$C_{\text{lin}} = C_{\text{red}}C_\xi + c_{\Omega,0}C_{\text{sym}}C_{\mathfrak{L}_\xi\xi}\gamma\delta^\tau$	(5.32)
$\bar{E}$	$C_E = 2(C_L + C_N)C_{\text{lin}}\gamma\delta^{\tau-1} + \frac{1}{2}c_{Z_h,2}(C_{\Delta K})^2$	(5.36)
ΔL	$C_{\Delta L} = nC_{\Delta K}$	(5.39)
$\Delta L^\top$	$C_{\Delta L^\top} = 2nC_{\Delta K}$	(5.40)
ΔG	$C_{\Delta G} = c_{G,1}C_{\Delta K}$	(5.41)
ΔG_L	$C_{\Delta G_L} = C_{L^\top}c_{G,0}C_{\Delta L} + C_{L^\top}C_{\Delta G}C_L\delta + C_{\Delta L^\top}c_{G,0}C_L$	(5.42)
ΔB	$C_{\Delta B} = (\sigma_B)^2 C_{\Delta G_L}$	(5.44)
$\Delta\Omega$	$C_{\Delta\Omega} = c_{\Omega,1}C_{\Delta K}$	(5.45)
ΔJ	$C_{\Delta J} = c_{J,1}C_{\Delta K}$	(5.46)
$\Delta J^\top$	$C_{\Delta J^\top} = c_{J^\top,1}C_{\Delta K}$	(5.47)
ΔN^0	$C_{\Delta N^0} = c_{J,0}C_{\Delta L} + C_{\Delta J}C_L\delta$	(5.48)
$\Delta N^{0\top}$	$C_{\Delta N^{0\top}} = C_{\Delta L^\top}c_{J^\top,0} + C_{L^\top}C_{\Delta J^\top}\delta$	(5.49)
$\Delta\bar{N}$	$C_{\Delta\bar{N}} = C_{\Delta N^0}\sigma_B + C_{N^0}C_{\Delta B}$	(5.50)
$\Delta\bar{N}^\top$	$C_{\Delta\bar{N}^\top} = \sigma_B C_{\Delta N^{0\top}} + C_{\Delta B}C_{N^0,\top}$	(5.51)
ΔA	$C_{\Delta A} = \frac{1}{2}\left(C_{\Delta\bar{N}^\top}c_{\Omega,0}C_{\bar{N}} + C_{\bar{N}^\top}C_{\Delta\Omega}C_{\bar{N}}\delta + C_{\bar{N}^\top}c_{\Omega,0}C_{\Delta\bar{N}}\right)$ $^*C_{\Delta A} = 0$	(5.52)
ΔN	$C_{\Delta N} = C_A C_{\Delta L^\top} + C_{\Delta A}C_{L^\top} + C_{\Delta\bar{N}}$	(5.53)
$\Delta N^\top$	$C_{\Delta N^\top} = C_A C_{\Delta L^\top} + C_{\Delta A}C_{L^\top} + C_{\Delta\bar{N}^\top}$	(5.54)
ΔT_h	$C_{\Delta T_h} = c_{T_h,1}C_{\Delta K}$	(5.57)
$\Delta T_h^\top$	$C_{\Delta T_h^\top} = c_{T_h^\top,1}C_{\Delta K}$	(5.58)
ΔT	$C_{\Delta T} = \frac{1}{2}C_{\Delta\bar{N}^\top}c_{T_h,0}C_{\bar{N}} + C_{\bar{N}^\top}C_{\Delta T_h}C_{\bar{N}}\delta + C_{\bar{N}^\top}c_{T_h,0}C_{\Delta\bar{N}}$ $+ \frac{1}{2}C_{\Delta\bar{N}^\top}c_{T_h^\top,0}C_{\bar{N}} + C_{\bar{N}^\top}C_{\Delta T_h^\top}C_{\bar{N}}\delta + C_{\bar{N}^\top}c_{T_h^\top,0}C_{\Delta\bar{N}}$ $+ C_{\Delta\bar{N}^\top}c_{T_h^\top,0}C_N + C_{\bar{N}^\top}C_{\Delta T_h^\top}C_N\delta + C_{\bar{N}^\top}c_{T_h^\top,0}C_{\Delta N}$ $+ C_{\Delta N^\top}c_{T_h,0}C_{\bar{N}} + C_{N^\top}C_{\Delta T_h}C_{\bar{N}}\delta + C_{N^\top}c_{T_h,0}C_{\Delta\bar{N}}$	(5.59)
$\Delta\langle T\rangle^{-1}$	$C_{\Delta\langle T\rangle^{-1}} = (\sigma_T)^2 C_{\Delta T}$	(5.61)

Table 3: Constants introduced in Section 5.3 associated with the convergence of que quasi-Newton method that yields Theorem 3.1, i.e., the ordinary formulation.

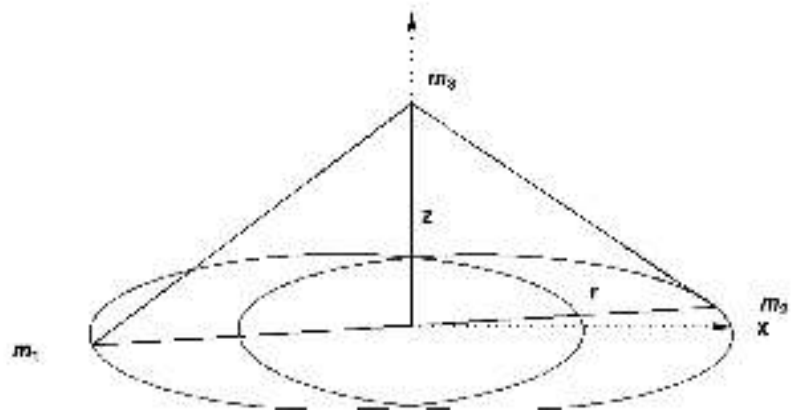
Constant	Label
$\mathfrak{C}_1 = \max\left\{(a_1a_3)^{4\tau}C_E, (a_3)^{2\tau+1}\gamma^2\rho^{2\tau-1}C_\Delta\right\}$	(5.67)
$C_{\Delta,1} = \max\left\{\frac{nC_{\Delta K}}{\sigma_L - \|\mathrm{D}K\|_\rho}, \frac{2nC_{\Delta K}}{\sigma_{L^\top} - \|\mathrm{D}K^\top\|_\rho}, \frac{C_{\Delta B}}{\sigma_B - \|B\|_\rho}, \frac{C_{\Delta\langle T\rangle^{-1}}}{\sigma_T - \|\langle T\rangle^{-1}\|}\right\}$	(5.68)
$C_{\Delta,2} = \frac{C_{\Delta K}\delta}{\text{dist}(K(\mathbb{T}^d_\rho),\partial\mathcal{B})}$	(5.69)
$C_\Delta = \max\left\{C_{\text{sym}}\gamma\delta^\tau, \frac{C_{\Delta,1}}{1-a_1^{1-2\tau}}, \frac{C_{\Delta,2}}{1-a_1^{-2\tau}}\right\}$	(5.70)
$\mathfrak{C}_2 = a_3^{2\tau}C_{\Delta K}/(1-a_1^{-2\tau})$	(5.71)

The Sitnikov Problem

Let m_1, m_2, m_3 be the masses of three point particles P_1, P_2, P_3, with P_1, P_2 the primaries and P_3 the satellite.

Three classical hypotheses:

1. Primaries have equal mass $m_1 = m_2 \equiv m$ and move in the plane $Z = 0$ on antisymmetric Keplerian ellipses with common focus at the origin
2. Third body is massless $(m_3 = 0)$
3. Satellite starts on the Z–axis with zero planar components



Sitnikov configuration: primaries in plane, satellite on Z-axis

Under these assumptions, the satellite's vertical motion decouples, yielding a time-periodic one-degree-of-freedom Hamiltonian.

The Sitnikov Hamiltonian (Work in progress)

If z denotes the vertical coordinate of the satellite and p_z its conjugate momentum, the time-dependent Sitnikov Hamiltonian reads

$$H_e(z, p_z, t) = \frac{1}{2}p_z^2 - \frac{1}{\sqrt{z^2 + \big(a(1 - e\cos E(t))\big)^2}},$$

where the eccentric anomaly $E(t)$ satisfies **Kepler's equation**:

$$E(t) - e\sin E(t) = \frac{t - t_0}{a^{3/2}}.$$

For $e = 0$ (circular case), $E(t)$ is linear in t; for $e > 0$, the motion is slower at apoapsis and faster at periapsis.

The MacMillan Hamiltonian (Circular Case)

For the circular case $e = 0$, the system is **autonomous**. We obtain the MacMillan Hamiltonian

$$H_0(z, p_z) = \frac{1}{2}p_z^2 \; - \; \frac{1}{\sqrt{z^2 + a^2}}, \qquad a > 0, \tag{3}$$

which defines an **integrable** one–degree–of–freedom system.

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which defines an **integrable** one–degree–of–freedom system.

The period is given by

$$\tau = \frac{4a^{\frac{3}{2}}}{(1-2k^2)}\left[E(k) + \frac{\pi}{2\sqrt{2(1-2k^2)}}\left(1 - \Lambda_0\left(\arcsin\sqrt{\tfrac{1-2k^2}{1-k^2}}\right)\right)\right], \tag{4}$$

where $k^2 = \frac{1+ah}{2}$, and $E(k)$ is the **complete elliptic integral of the second kind**.

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where $k^2 = \frac{1+ah}{2}$, and $E(k)$ is the **complete elliptic integral of the second kind**.

The function $\Lambda_0(\varphi, k)$ is the **Heuman Lambda function**:

$$\Lambda_0(\varphi, k) = \frac{2(1-k^2)\sin\varphi\cos\varphi}{\pi\sqrt{1-(1-k^2)\sin^2\varphi}}\int_0^{K(k)}\left[1 - \frac{k^2\,\mathrm{sn}^2 u}{1-(1-k^2)\sin^2\varphi}\right]^{-1} du,$$

where $K(k)$ is the complete elliptic integral of the first kind, and $\mathrm{sn}(u, k)$ is the Jacobi elliptic sine.

Initial Guess: Setup

Since the eccentric anomaly E is 2π–periodic, we introduce

$$\varphi \in \mathbb{T} \quad \text{defined by} \quad 2\pi\varphi = E.$$

The angular variable θ is then defined as

$$\theta = \frac{a^{3/2}}{\tau} E = 2\pi a^{3/2} \frac{\varphi}{\tau}.$$

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In these coordinates, the system becomes

$$\begin{cases} \dfrac{dz}{d\theta} = \tau\big(1 - e\cos(2\pi\varphi)\big)\, p_z, \\ \dfrac{dp_z}{d\theta} = -\dfrac{\tau\big(1 - e\cos(2\pi\varphi)\big)\, z}{\Big(z^2 + \big(a(1 - e\cos(2\pi\varphi))\big)^2\Big)^{3/2}}, \\ \dfrac{d\varphi}{d\theta} = \dfrac{\tau}{2\pi a^{3/2}}. \end{cases}$$

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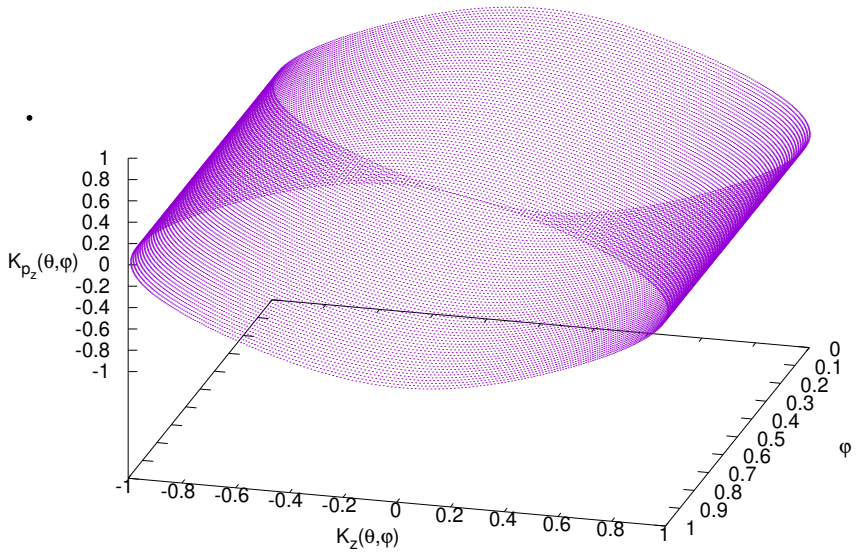
The frequency vector is $(\omega, \alpha) = \left(2\pi,\ \frac{\tau}{2\pi a^{3/2}}\right)$.

Initial Torus Parameterization

We parametrize the torus by

$$\begin{pmatrix} K(\theta,\varphi) \\ \varphi \end{pmatrix} = \begin{pmatrix} K_z(\theta) \\ K_{p_z}(\theta) \\ \varphi \end{pmatrix} = \begin{pmatrix} z(\theta) \\ p_z(\theta) \\ \varphi \end{pmatrix}.$$

In the integrable case ($e = 0$), φ is constant along the torus; the dynamics is confined to the θ-direction.



KAM Iteration Results (e=0)

- Frequency vector: $\omega = (2\pi, 1.7322255908)$
- Fourier grid: 512×64 (32768 modes)
- Extended precision (68-bit arithmetic)

Iteration	Invariance Error
0	1.52×10^{-1}
1	9.52×10^{-3}
2	1.96×10^{-5}
3	3.53×10^{-11}
4	3.99×10^{-16}

Quadratic convergence observed.

- Newton correction decreased from 2.28×10^{-2} to 8.33×10^{-13}
- Torsion matrix: $T(\theta, \varphi) \approx -8.0007$, $T^{-1} \approx -1.25 \times 10^{-1}$
- Twist non-degeneracy condition verified

This torus was used as a numerical seed for continuation in the perturbation parameter e.

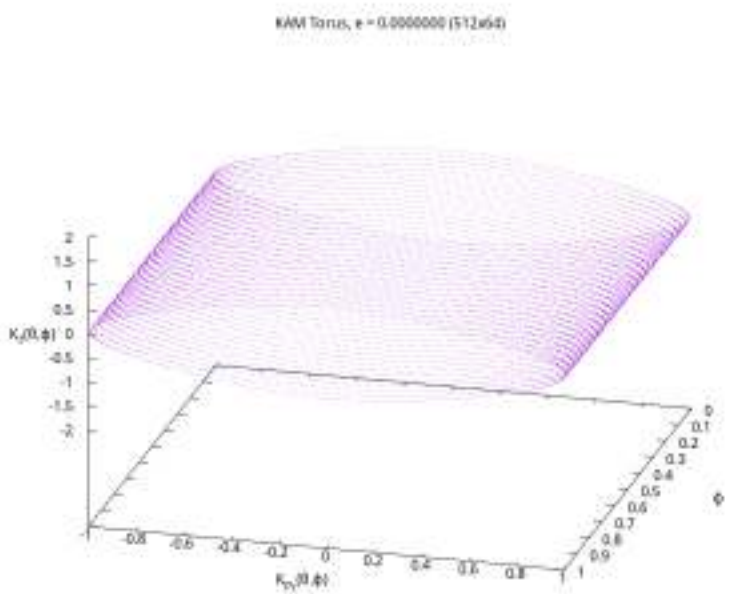
Continuation algorithm

We can continue the family of tori with respect to the parameter ϵ.

1. Construct an approximately invariant parametrization.
2. Improve the parametrization $K_\epsilon(\theta, \varphi)$ using the KAM algorithm.
3. Save the updated torus $K_\epsilon(\theta, \varphi)$.
4. Define a continuation step δ.
5. Compute $K_{\epsilon+\delta\epsilon}(\theta, \varphi)$ using the KAM algorithm.
6. Repeat from step (2) until the desired value of ϵ is reached.

Continuation of the Invariant Torus with Eccentricity

- Evolution from $e = 0$ to $e = 0.514$
- 59 invariant tori computed
- Grid refinement:
  - $e \leq 0.06$: 512×64
  - $e \leq 0.24$: 512×128
  - $e \leq 0.46$: 512×256
  - $e \leq 0.514$: 512×512



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A Tokamak Model

We apply our results to a Tokamak using the model proposed by Chandre et al. (2000).

$$H(\theta,\psi;\varphi)=\int \frac{d\psi}{q(\psi)}+\epsilon H_1(\theta,\psi,\varphi)+\epsilon^2 f_2(\theta,\varphi),$$

where

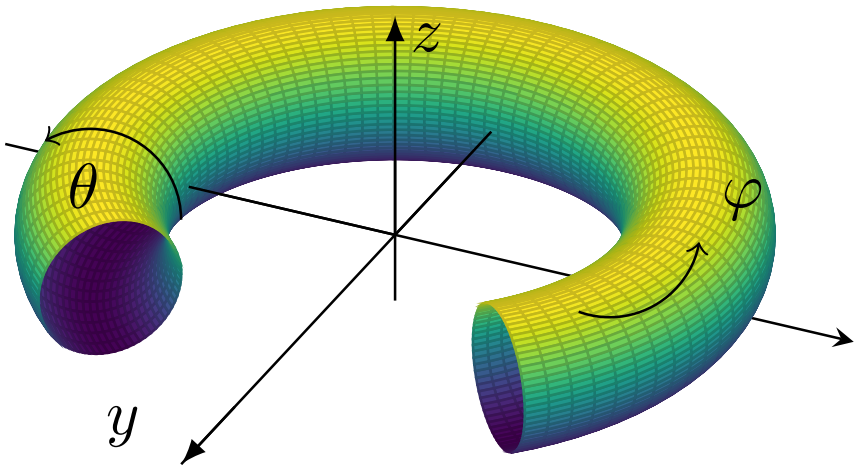
$$q(\psi)=\frac{4}{(2-\psi)(2-2\psi+\psi^2)},$$

$$H_1(\theta,\varphi)=\cos(2\theta-\varphi)+\cos(3\theta-2\varphi),$$

$$f_2(\theta,\varphi)=\frac{5427}{3200}\left(\frac{2\cos(2\theta-\psi)}{2\omega-1}+\frac{3\cos(3\theta-2\psi)}{3\omega-2}\right)^2,$$

$$\omega=\frac{1}{q(0.35)},$$

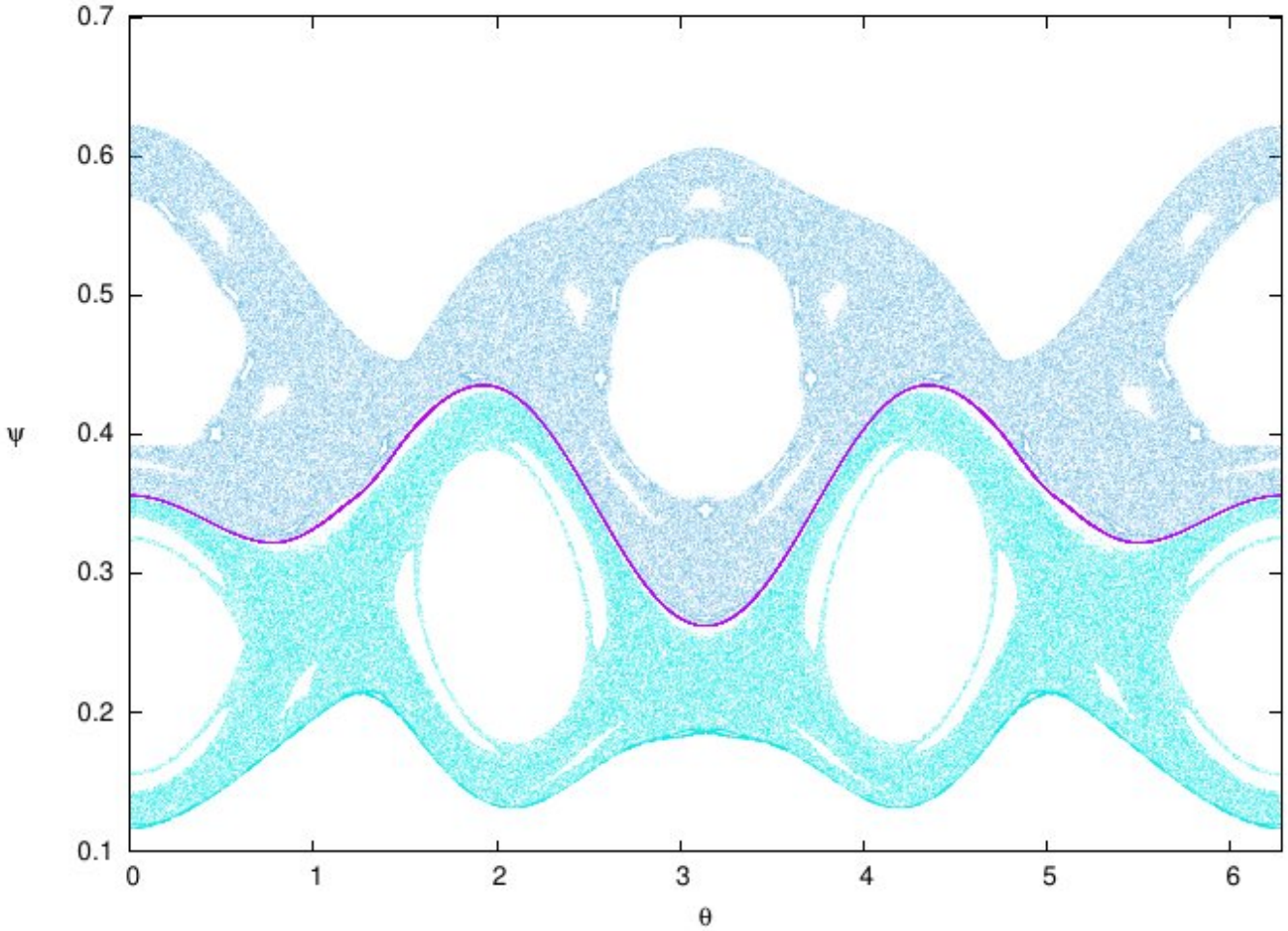
$$\epsilon=0.004.$$



Where θ is the poloidal angle, ψ is its conjugate variable and φ is the toroidal angle, which plays the role of time.

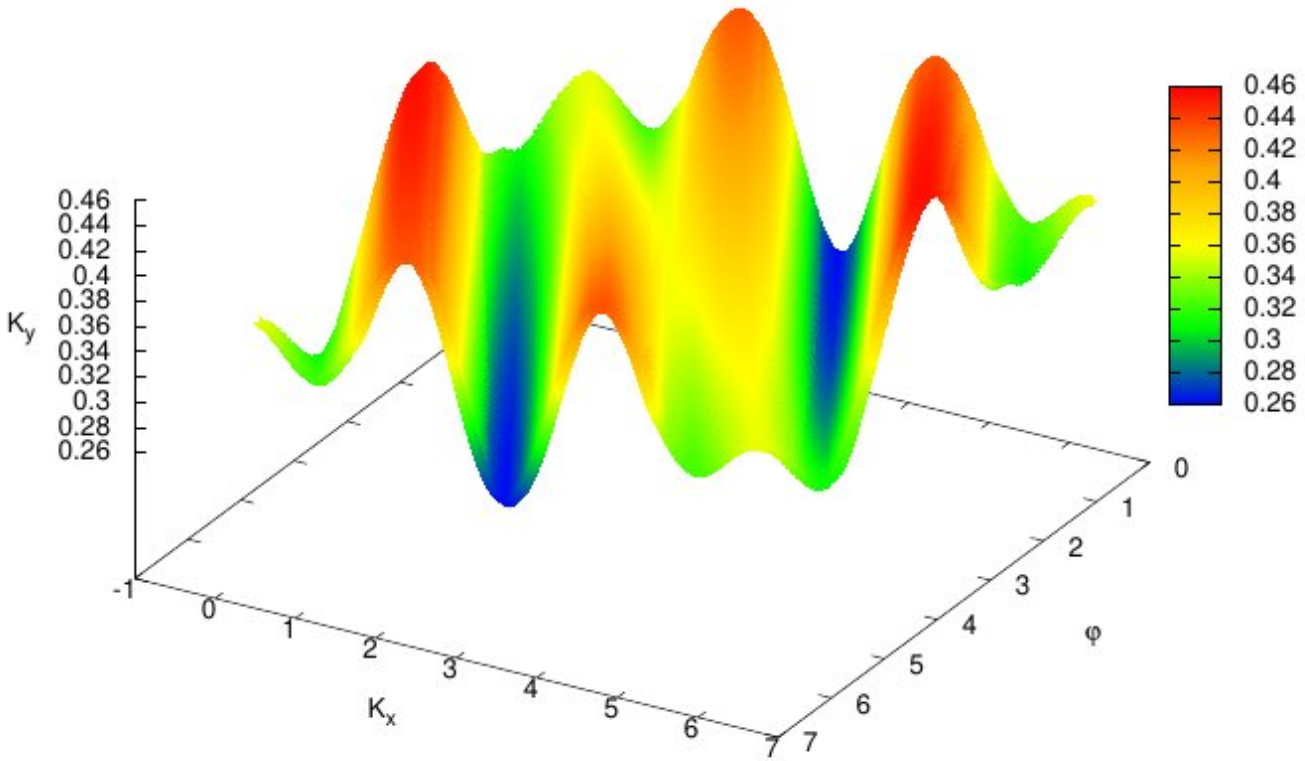
Stroboscopic application

They introduce a control method to confine the plasma between chaotic regions.



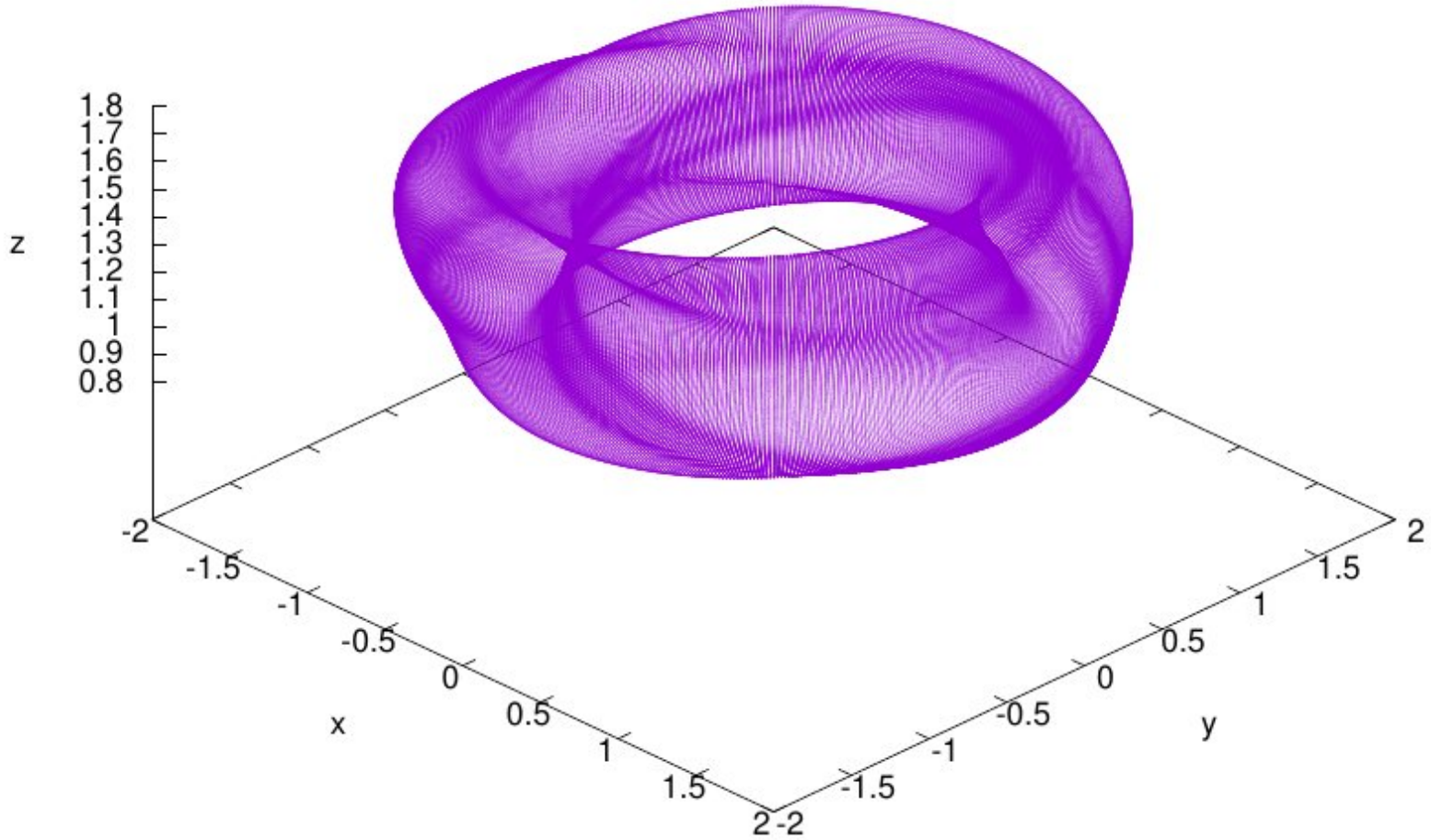
Invariant torus for the tokamak

This result was published in Calleja, Haro & Porras (2026). The initial invariance error is 0.0968657. With quasi-Newton method we obtain an approximately invariant torus with an error below 4×10^{-16}.



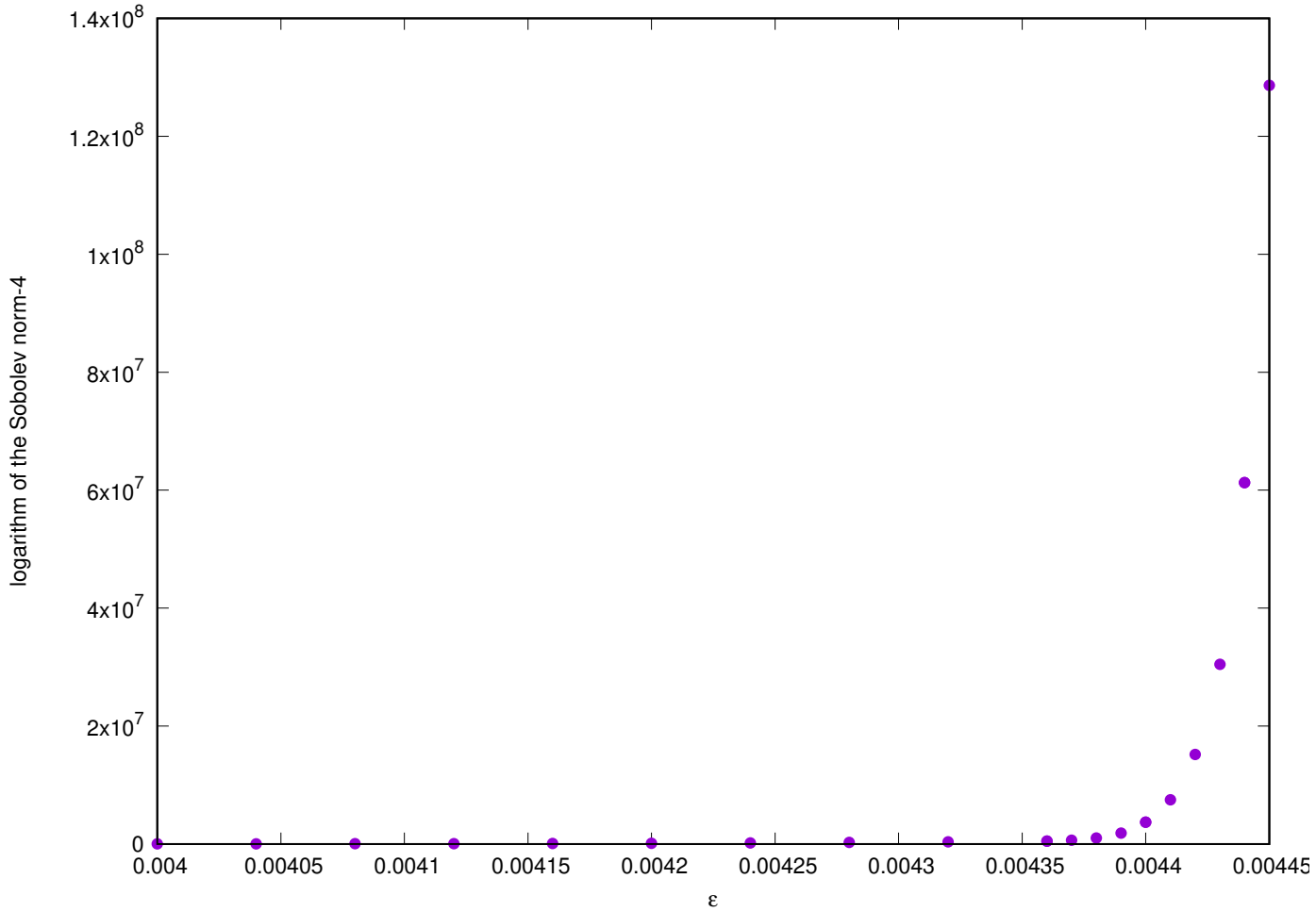
Invariant torus for the tokamak

This torus was parameterized using $2^{11} \times 2^{11}$ Fourier coefficients in θ and φ, respectively.



Continuation of the invariant torus

The Sobolev norm exhibits exponential growth near the breakdown threshold.



Sobolev Norm as a Diagnostic of Breakdown

Scale law. Following Calleja & de la Llave (2010), the breakdown of invariant tori can be estimated via the scale law

$$\|K\|_r = \frac{b}{(\varepsilon - \varepsilon_c)^a}.$$

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Linear fit. The logarithm of the Sobolev H^r norm is linearly fitted as

$$\log\big(\|K\|_r\big) = -a(r)\,\log\big(\varepsilon - \varepsilon_c\big) + b(r), \qquad \text{with } \varepsilon_c \approx 0.004522.$$

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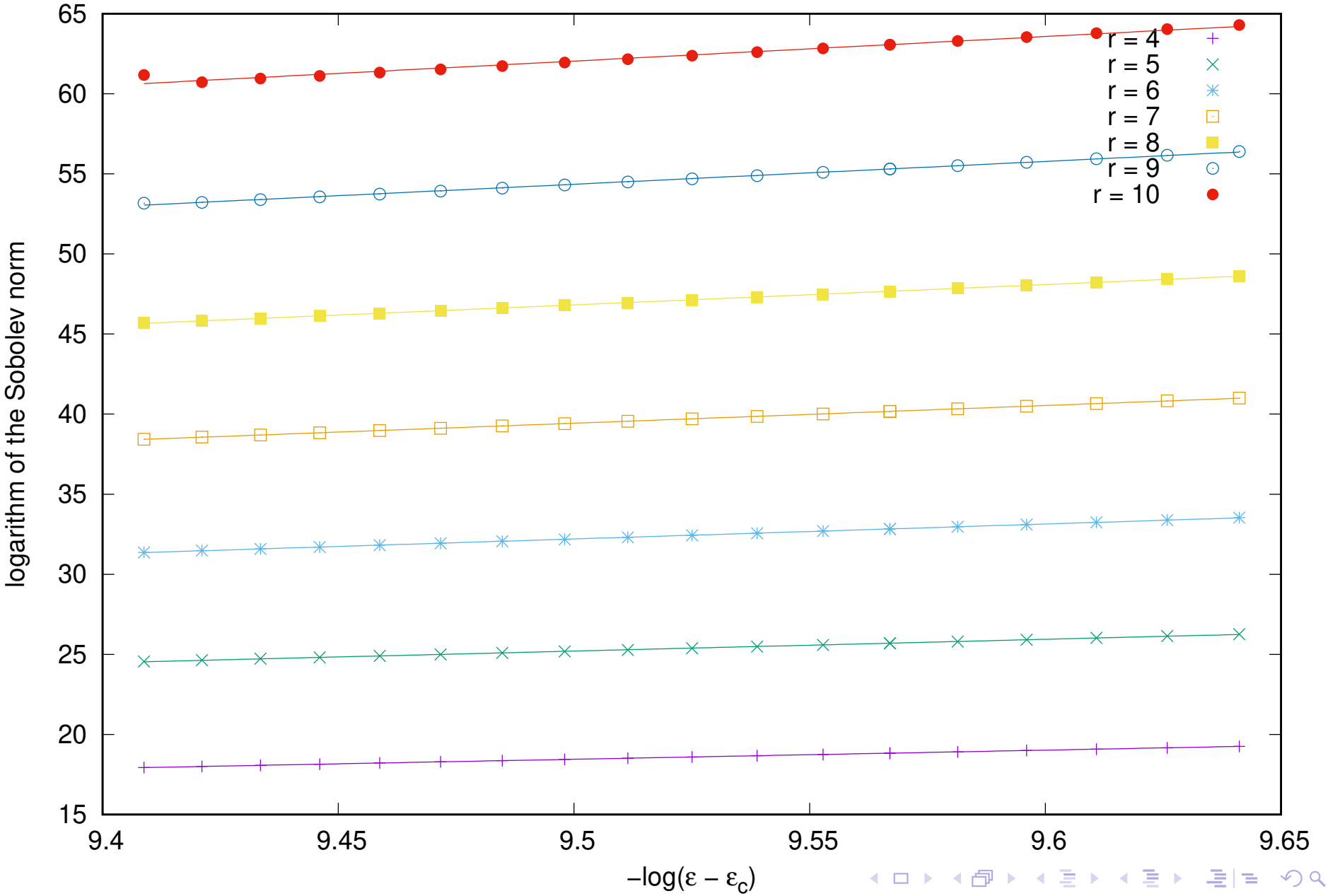
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r	$a(r)$	$b(r)$	Correlation
4	5.691210	-35.614620	0.999997
5	7.348686	-44.605575	0.999826
6	9.337807	-56.503079	0.999952
7	11.045949	-65.505097	0.999984
8	12.674821	-73.591227	0.999944
9	14.262883	-81.147597	0.999479
10	15.341002	-83.699312	0.992078

Linear Regression for $\|K\|_r$



Quasi-periodic Pendulum

Consider the Hamiltonian

$$H(q,p,t) = \frac{p^2}{2} + \epsilon_1 \cos(q) - \big(\epsilon_2 + \epsilon_3 \cos(\varphi_1)\big) \cos(q - \varphi_2), \tag{5}$$

where $\varphi_1 = \sqrt{3}\, t$ and $\varphi_2 = \frac{1+\sqrt{5}}{2}\, t$. The external frequencies are $\alpha_1 = \sqrt{3}$ and $\alpha_2 = \frac{1+\sqrt{5}}{2}$.

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1. **Single wave model:** $\epsilon_1 = 0$.
   We start from the planar pendulum as an initial approximation ($\epsilon_2 = \epsilon_3 = 0$) and perform continuation by gradually increasing ϵ_2 and ϵ_3. This model serves as a simplification of the vorticity mixing problem.

Quasi-periodic Pendulum

Consider the Hamiltonian

$$H(q,p,t) = \frac{p^2}{2} + \epsilon_1 \cos(q) - \big(\epsilon_2 + \epsilon_3 \cos(\varphi_1)\big) \cos(q - \varphi_2), \tag{5}$$

where $\varphi_1 = \sqrt{3}\, t$ and $\varphi_2 = \frac{1+\sqrt{5}}{2}\, t$. The external frequencies are $\alpha_1 = \sqrt{3}$ and $\alpha_2 = \frac{1+\sqrt{5}}{2}$.

1. **Single wave model:** $\epsilon_1 = 0$.
   We start from the planar pendulum as an initial approximation ($\epsilon_2 = \epsilon_3 = 0$) and perform continuation by gradually increasing ϵ_2 and ϵ_3. This model serves as a simplification of the vorticity mixing problem.
2. **Quasi-periodic pendulum:** $\epsilon_1 = 1$
   Starting from the simple pendulum ($\epsilon_2 = \epsilon_3 = 0$), we apply continuation with respect to (ϵ_2, ϵ_3), obtaining two families of invariant tori:
   - **Rotational tori.**
   - **Librational tori.**

   Both cases are treated with the same continuation scheme, although they exhibit different topological structures.

Single Wave Model

$$H(q,p,t) = \frac{p^2}{2} - \big(\epsilon_2 + \epsilon_3 \cos(\varphi_1)\big) \cos(q - \varphi_2),$$

Parameters: $\epsilon_2 = 0.320$, $\epsilon_3 = 0.155$, and internal frequency $\omega = 2$.

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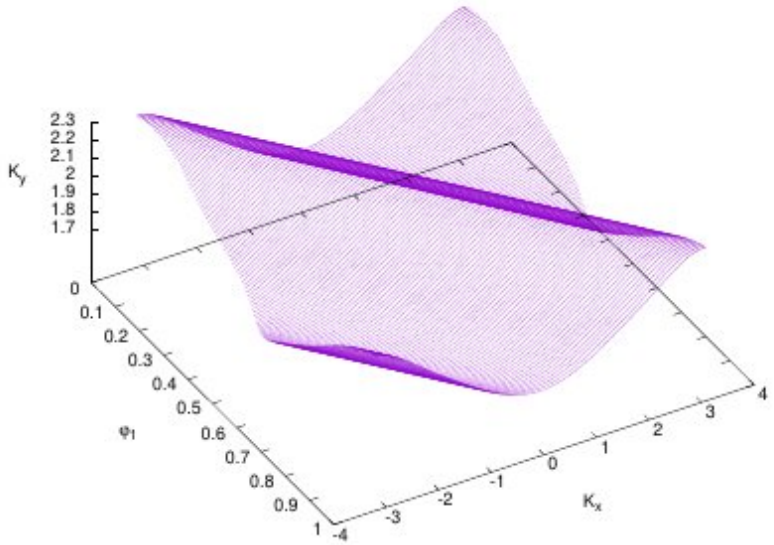


Figure: Torus projection (direction 1).

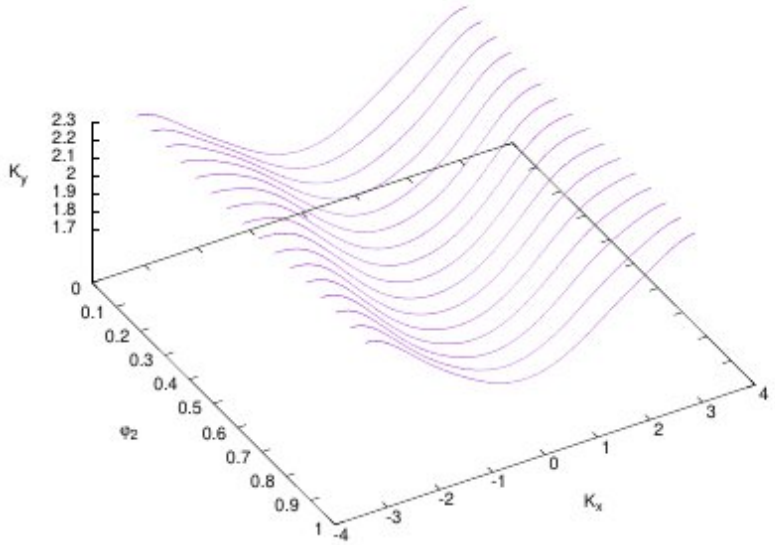


Figure: Torus projection (direction 2).

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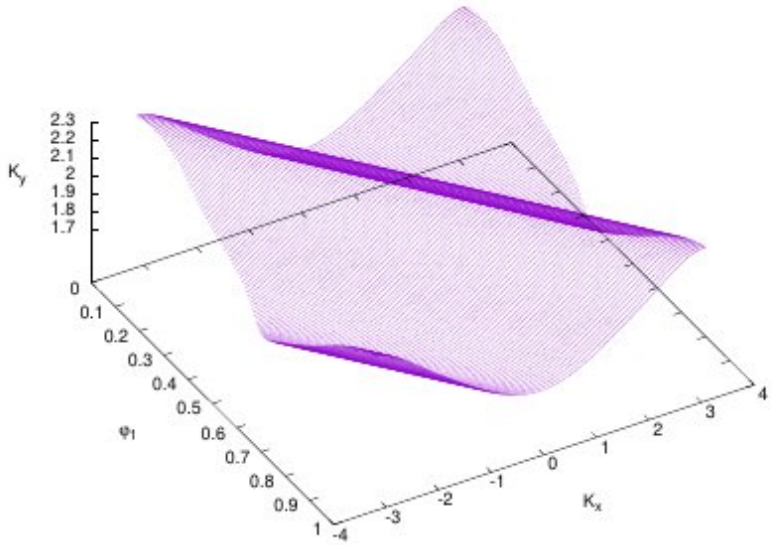


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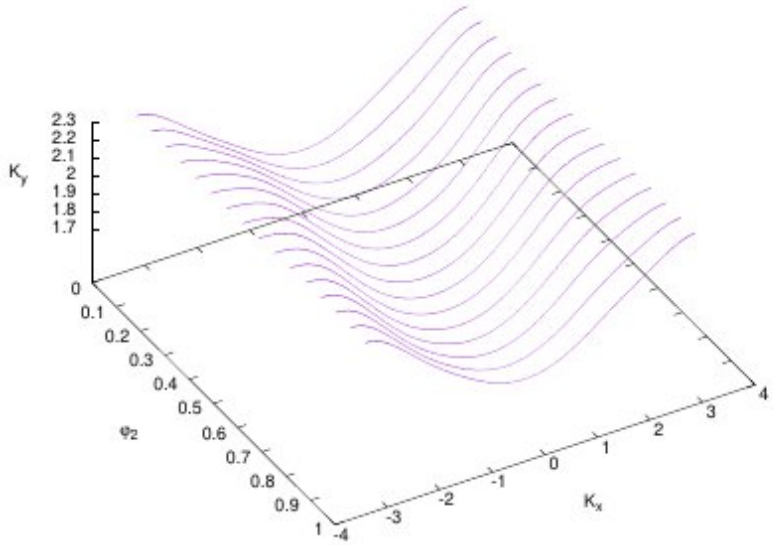


Figure: Torus projection (direction 2).

The number of Fourier coefficients in each direction is $2^9 \times 2^9 \times 2^7 = 2^{25}$ for θ, φ_1, and φ_2, respectively.

Rotational Motions

$$H(q,p,t) = \frac{p^2}{2} + \cos(q) - \big(\epsilon_2 + \epsilon_3 \cos(\varphi_1)\big) \cos(q - \varphi_2),$$

Parameters: $\epsilon_2 = 0.080$, $\epsilon_3 = 0.040$, and internal frequency

$$\omega = 2.395097499769847499 99.$$

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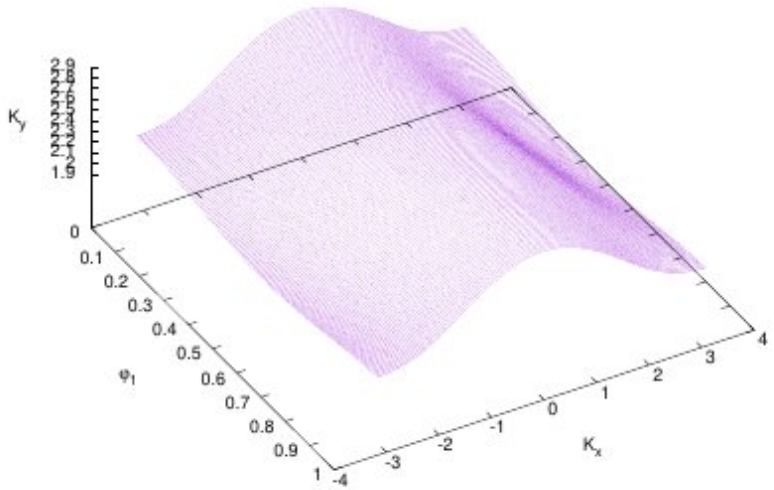


Figure: Rotational torus (direction 1).

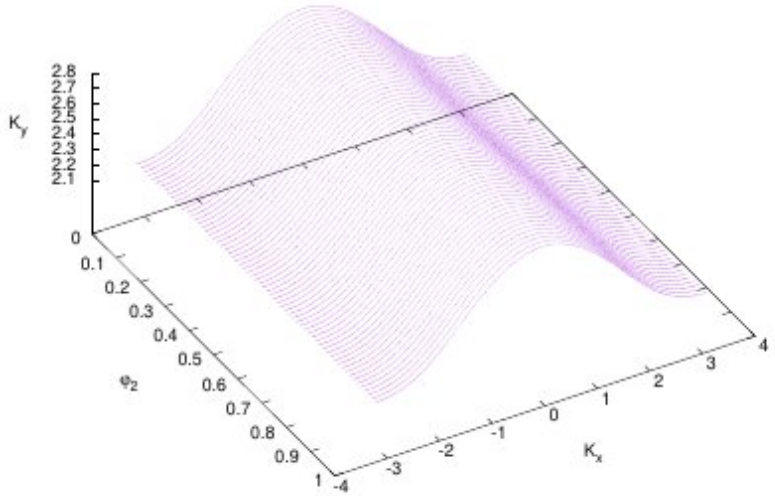


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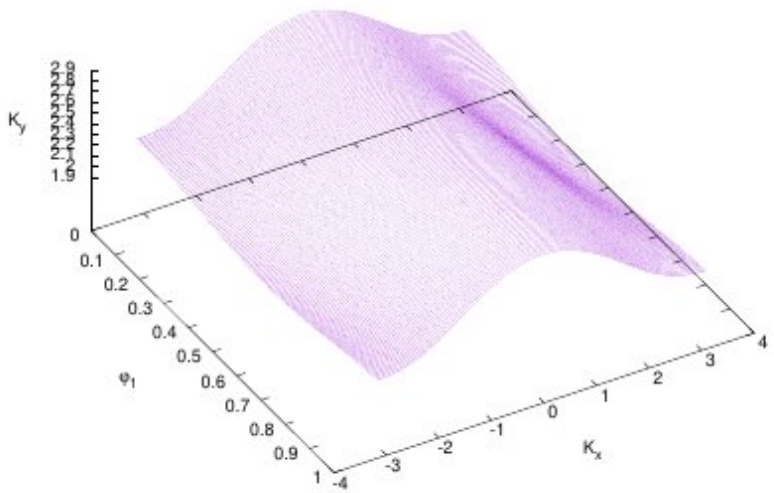


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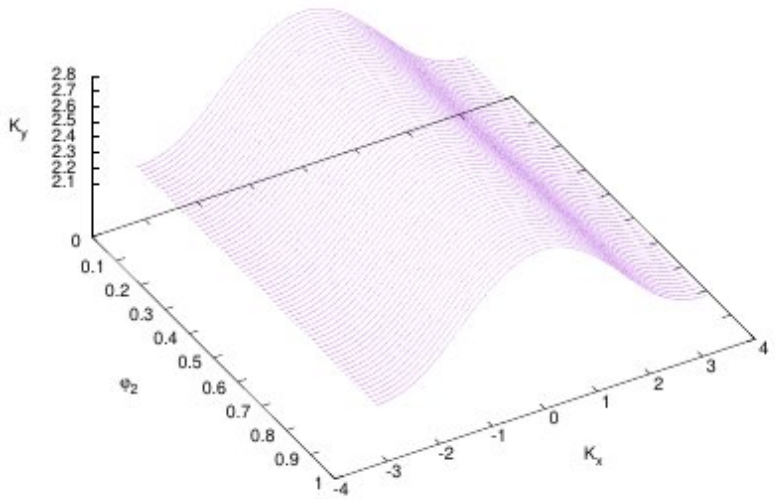


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Parameters: $\epsilon_2 = 0.070$, $\epsilon_3 = 0.014$, and internal frequency

$$\omega = 0.84721308479397908659.$$

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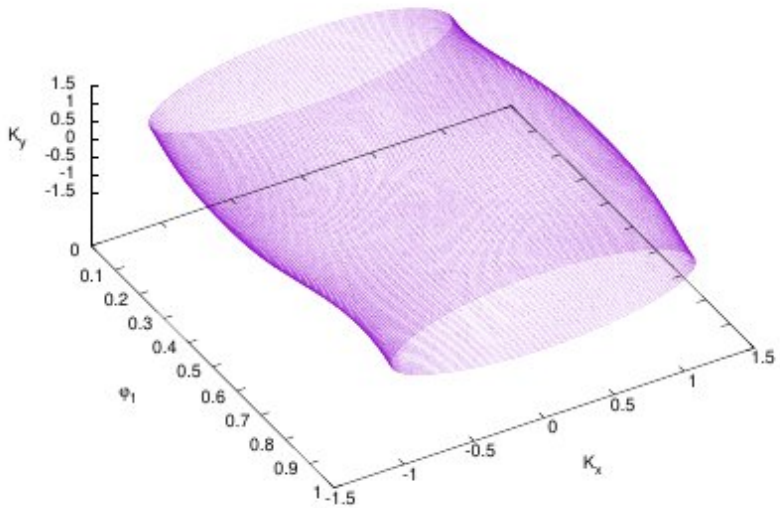


Figure: Librational torus (direction 1).

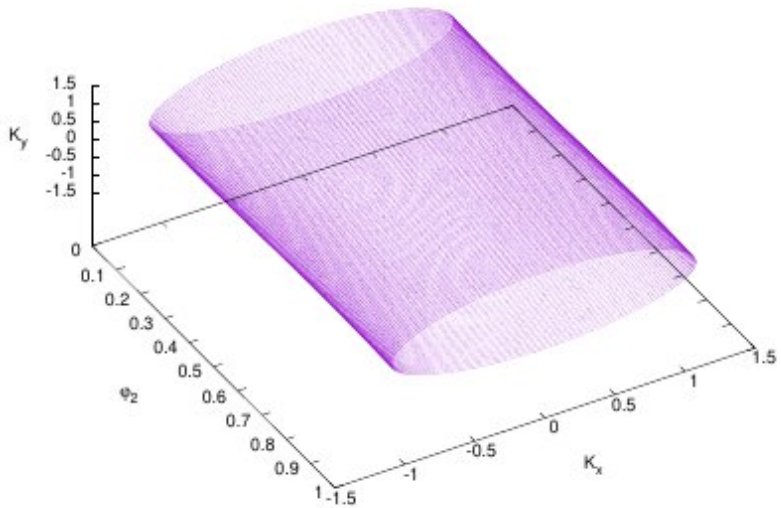


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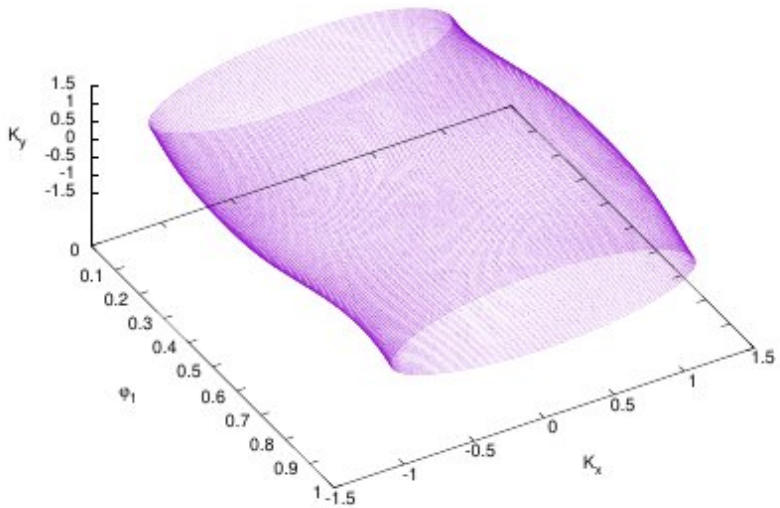


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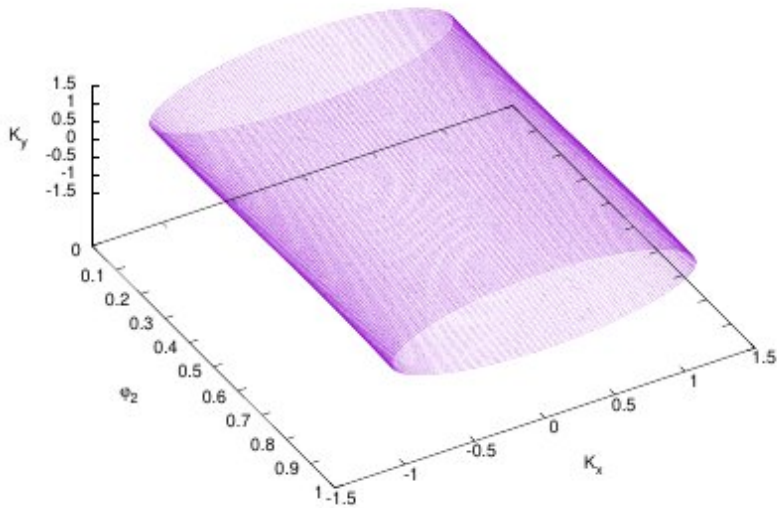


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Main Messages

- We start from an approximately invariant manifold and improve it by adding a correction.
- By iterating the process, the method converges toward the truly invariant torus.
- The schemes derived from the proof can be turned into efficient and reliable numerical methods for computing invariant tori.
- We are working on the validation of the numerical computations to produce **Computer-Assisted Proofs**.

Thank you for your attention

General scheme of a computer-assisted proof

1. **Initial numerical computation:** obtain an approximate solution $K(\theta, \varphi)$ of the invariant torus with adequate precision.
2. **Rigorous computation 1:** estimate control constants $\mathfrak{C}_1$, γ, τ and the analyticity radius ρ using *interval arithmetic* and decay of Fourier coefficients.
3. **Invariance error:** compute

$$E(\theta, \varphi) = Z_h(K(\theta, \varphi), \varphi) + \mathfrak{L}_{\omega,\alpha} K(\theta, \varphi),$$

   and its norm $\|E\|_\rho$ in the corresponding analytic space.
4. **Validation condition:** verify that

$$\frac{\mathfrak{C}_1 \|E\|_\rho}{\gamma^4 \rho^{4\tau}} < 1.$$

   If satisfied, the torus K is **rigorously invariant**.
5. **Conclusion:** the numerical result becomes a **computer-assisted mathematical proof**.

"The proof consists of verifying a quantitative inequality that guarantees the existence of the invariant torus."

Error estimates for analytic periodic functions

- To rigorously validate approximate solutions, it is necessary to control the error between the analytic function f and its discrete approximation $\tilde{f}$ on a finite Fourier grid. This control allows bounding the norm $\|E\|_\rho$ that appears in the validation condition:

$$\frac{\mathfrak{C}_1 \|E\|_\rho}{\gamma^4 \rho^{4\tau}} < 1.$$

- **Context:** f is analytic in a complex strip $\mathbb{T}^n_{\hat{\rho}}$, and $\tilde{f}$ is its discrete projection on a grid of size N_F.

- **Key result (Theorem 3.3) Haro & Figueras 2019:**

$$\|\tilde{f} - f\|_\rho \leq C_{N_F}(\rho, \hat{\rho}) \|f\|_{\hat{\rho}}, \quad 0 \leq \rho < \hat{\rho},$$

where $C_{N_F}(\rho, \hat{\rho})$ is obtained explicitly from the constants $S^{*1}_{N_F}$, $S^{*2}_{N_F}$ and T_{N_F}, which control the exponential decay of the truncated modes.

Sobolev norm as a breakdown diagnostic

Let $r \in \mathbb{R}^+$, the Sobolev space H^r is a Banach space formed by functions K, characterized by the norm

$$\|K\|_r = \sqrt{\sum_{k_1 \in \mathbb{Z}^n} \sum_{k_2 \in \mathbb{Z}^\ell} |2\pi i(k_1 + k_2)|^{2r} |\hat{v}_{k_1,k_2}|^2} < \infty.$$

Here $|\cdot|$ denotes the maximum norm in $\mathbb{R}^n$ or $\mathbb{C}^n$:
To analyze the breakdown threshold, we consider the Sobolev H^4 norm of K, defined as

$$\|K\|_4 := \sqrt{\|K_x^p\|_4^2 + \|K_y^p\|_4^2}\,.$$

A rapid growth of this norm indicates the onset of breakdown, linking the phenomenon with the regularity properties of the system.

Cohomological equations

Consider a periodic function

$$v : \mathbb{T}^n \times \mathbb{T}^\ell \to \mathbb{R}, \qquad (\omega, \alpha) \in \mathbb{R}^n \times \mathbb{R}^\ell.$$

Its Fourier expansion is

$$v(\theta, \varphi) = \sum_{k_1 \in \mathbb{Z}^n} \sum_{k_2 \in \mathbb{Z}^\ell} \hat{v}_{k_1,k_2} \, e^{2\pi i (k_1 \cdot \theta + k_2 \cdot \varphi)},$$

with coefficients

$$\hat{v}_{k_1,k_2} = \int_{\mathbb{T}^\ell} \int_{\mathbb{T}^n} v(\theta, \varphi) e^{-2\pi i (k_1 \cdot \theta + k_2 \cdot \varphi)} d\theta \, d\varphi.$$

We define the average as

$$\langle v \rangle := \hat{v}_{0,0}.$$

Spectral profiles

To separate the contribution of each direction, we write

$$\theta = (\theta_1, \dots, \theta_n), \qquad \varphi = (\varphi_1, \dots, \varphi_\ell).$$

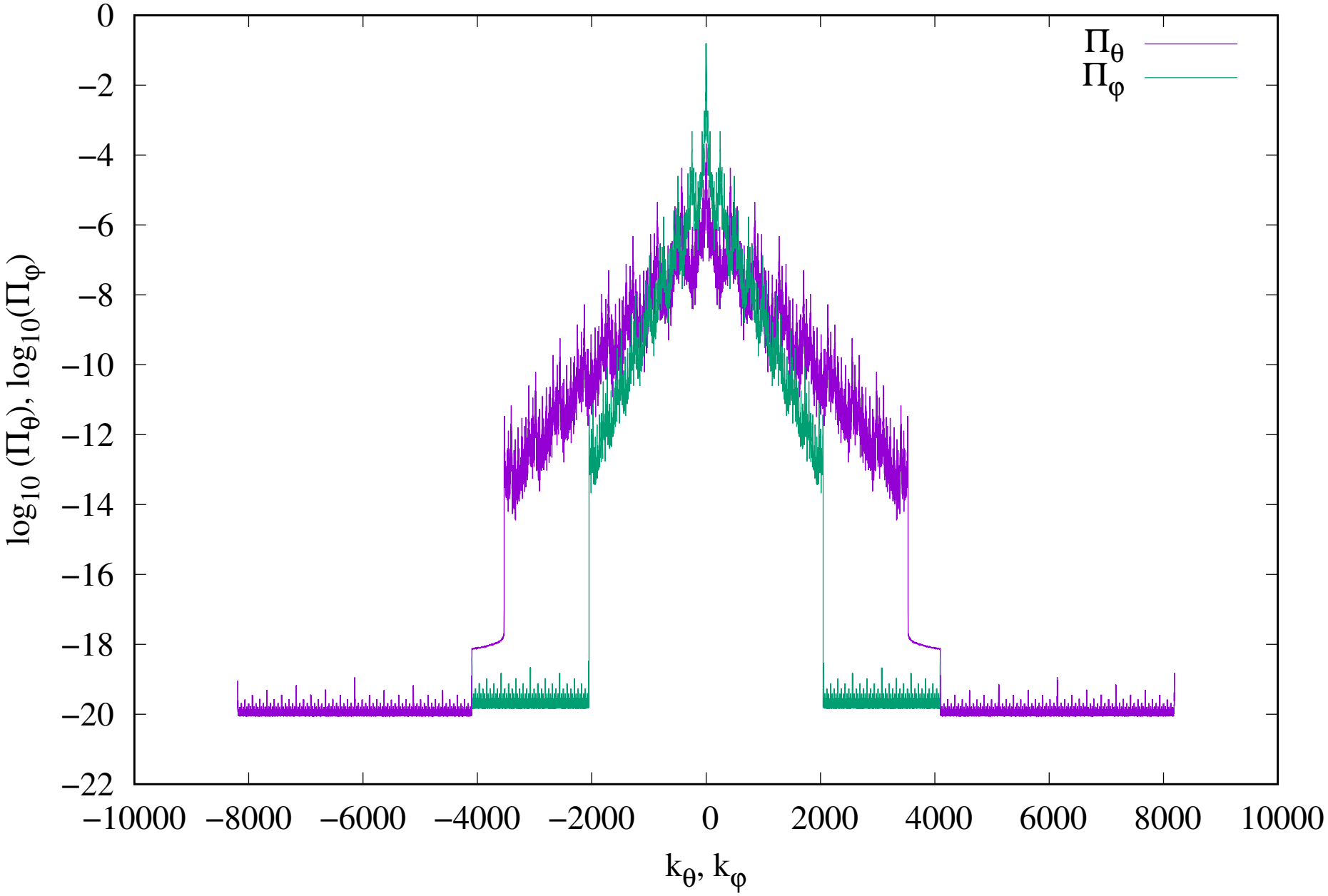
The spectral profiles are defined as:

$$\Pi_{\theta_j}(k_{\theta_j}) := \sum_{\substack{k_{\theta_m} \in \mathbb{Z} \\ m \neq j}} \sum_{k_2 \in \mathbb{Z}^\ell} \left| \hat{v}_{k_1, k_2} \right|, \quad j = 1, \dots, n,$$

$$\Pi_{\varphi_m}(k_{\varphi_m}) := \sum_{\substack{k_{\varphi_r} \in \mathbb{Z} \\ r \neq m}} \sum_{k_1 \in \mathbb{Z}^n} \left| \hat{v}_{k_1, k_2} \right|, \quad m = 1, \dots, \ell.$$

These profiles allow us to compare the effective analyticity widths in each direction.

Fourier modes of the torus



Cohomological equation

The cohomological equation is written as

$$\mathcal{L}_{\omega,\alpha} u = v - \langle v \rangle,$$

where $\mathcal{L}_{\omega,\alpha}$ is the Lie operator.
Its formal solution, with zero average, is

$$\mathcal{R}_{\omega,\alpha} v(\theta,\varphi) = \sum_{\substack{(k_1,k_2)\in\mathbb{Z}^n\times\mathbb{Z}^\ell \\ (k_1,k_2)\neq(0,0)}} \hat{u}_{k_1,k_2}\, e^{2\pi i(k_1\cdot\theta + k_2\cdot\varphi)},$$

with coefficients

$$\hat{u}_{k_1,k_2} = \frac{-\hat{v}_{k_1,k_2}}{2\pi i(k_1\cdot\omega + k_2\cdot\alpha)},$$

and $\hat{u}_{0,0}$ free.